LAKSHANA A

B.Tech Student, AI & Data Science • Software Developer

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INTERNSHIPS

National Informatics Centre (NIC), Govt. of India — *Jun 2026 – Jul 2026 (2 months)*
Backend Developer Intern (API Developer) | AI Oral Cancer Detection Application — On-site

- Developed 5 RESTful backend APIs (registration, login/authentication, time-series data) using **Python**, **Flask**, and **PostgreSQL** for an AI-powered healthcare application
- Designed and implemented a full authentication module (secure registration & login) with server-side request validation, resolving credential-validation and login errors through structured debugging
- Built a **Time-Series API** for sequential symptom-tracking data, enabling efficient structured storage and retrieval of patient history
- Tested and validated all APIs end-to-end using **Postman**, systematically covering registration, login, and time-series endpoints to catch edge cases and improve reliability before handoff
- Architected PostgreSQL schemas for structured relational storage of patient and session data, and maintained technical documentation for APIs and backend workflows within a government e-Governance environment

CodeAlpha — *Jan 2026 – Feb 2026*
Front-End Web Development Intern — Internship

- Developed responsive web pages using **HTML**, **CSS**, and **JavaScript**, implementing clean and structured UI components
- Improved website responsiveness across mobile, tablet, and desktop breakpoints through iterative cross-device testing and debugging
- Debugged and optimized front-end performance, following best coding practices and version control workflows

EDUCATION

Rajalakshmi Institute of Technology — *Chennai, TN*
B.Tech in Artificial Intelligence & Data Science — 2024 – 2028

- **CGPA:** 8.6/10
- Activities: Content Writer & Artist, Artist League (Cultural Club)

Rosary Matriculation Higher Secondary School — *Chennai, India*
Higher Secondary Education – 91.8% — 2023 – 2024

- Achievements: School First in Mathematics and English; Scored 100/100 in French – Higher Secondary (Class XII)

PROJECTS

HypeFiltr.ai | *JavaScript, Node.js, Express, Chrome Manifest V3, Gemini AI (OpenRouter)* | GitHub Demo — *Personal | May 2026*

- Built a solo AI-powered **Chrome extension** (Manifest V3) that scans webpages in real time to flag jargon, logical fallacies, bias, and unsourced claims, generating a live 0–100% Substance Score via a Bias Heatmap across 6 context profiles
- Deployed a **Node.js/Express** backend on Railway integrating **Gemini AI (OpenRouter)** with structured JSON prompting, Shadow DOM UI isolation, and persistent scan history via Chrome Storage API

Constellia | *Vanilla JS, HTML5, CSS3, MediaPipe* | GitHub Demo — *Personal | March 2026*

- Built a gesture-controlled learning platform using **MediaPipe** hand-tracking to transform study notes into interactive constellations, featuring spatial audio, a timed Speedrun Mode, and knowledge quizzes

IoT-Based Water-Borne Disease Detection Platform | *ESP32, MQTT, Firebase, Next.js, Node.js* | Demo — *Dec 2025*

- Built a real-time water quality monitoring system using **ESP32** sensors (pH, turbidity, temperature) streaming via **MQTT** to **Firebase**, with a **Next.js**/**Node.js** dashboard for live visualization and AI-driven contamination alerts
- Designed as a low-cost, scalable rural healthcare solution; selected in the **External Round of Smart India Hackathon 2025**

Hospital Emergency Prediction System | *Python, Scikit-learn, Streamlit, Pandas, NumPy* | GitHub — *Personal | November 2025*

- Built a **Linear Regression** model to predict hospital emergency admissions from historical data (prior admissions, temperature, holiday status, flu cases), with an interactive **Streamlit** dashboard for real-time prediction and resource planning

TECHNICAL SKILLS

Proficient in: Python, JavaScript, TypeScript, Node.js, SQL, Flask, PostgreSQL, HTML/CSS
Familiar with: Java, C/C++, React.js, MySQL, MongoDB, TensorFlow, Scikit-learn, Pandas, NumPy, Power BI
Core CS Competencies: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), Database Management Systems (DBMS), Operating Systems
Applied AI & Domains: LLM Integration (Gemini/OpenRouter), Computer Vision (MediaPipe), Chrome Extension APIs (Manifest V3), IoT (ESP32/MQTT), Git, Postman

HONORS & AWARDS

- **Selected in External Round**, Smart India Hackathon 2025
- Certified in **Artificial Intelligence** – Infosys Springboard
- Certified in **Data Structures and Algorithms** – Simplilearn (Credential ID: 9715495)
- Certified in **Ensemble Learning and Ensemble Learning Techniques** – Analytics Vidhya
- **National Service Scheme (NSS)** – 80+ hours of community service